

SETTLEMENT POINT, Bahamas

WMO: 994390

Lat: 26.7045N

Lon: 78.9947W

Elev: 2

StdP: 101.31

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 14.3       | 15.7 | 4.2                           | 5.1 | 15.0 | 5.8 | 5.7 | 16.1 | 20.5                  | 20.0 | 15.9 | 18.9 | 8.4                   | 300  | 0.489 |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 3.2                    | 31.2            | 26.8 | 30.6 | 26.2 | 30.0 | 25.8 | 27.6                | 31.0 | 27.1 | 30.4 | 26.7 | 29.8 | 4.2                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 26.7                            | 22.2 | 30.0 | 26.1 | 21.6 | 29.7 | 25.7 | 20.9 | 29.3 | 87.4          | 31.1 | 85.3 | 30.4 | 83.2 | 30.0 | 29.5              |

Extreme Annual Design Conditions

| Extreme Annual WS |      |                    | Extreme Annual Temperature |           |      |            | n-Year Return Period Values of Extreme Temperature |            |      |            |      |     |      |     |      |  |
|-------------------|------|--------------------|----------------------------|-----------|------|------------|----------------------------------------------------|------------|------|------------|------|-----|------|-----|------|--|
| Mean              |      | Standard Deviation |                            | n=5 years |      | n=10 years |                                                    | n=20 years |      | n=50 years |      |     |      |     |      |  |
| 1%                | 2.5% | 5%                 | Min                        | Max       | Min  | Max        | Min                                                | Max        | Min  | Max        | Min  | Max |      |     |      |  |
| (a)               | (b)  | (c)                | (e)                        | (f)       | (g)  | (h)        | (i)                                                | (j)        | (k)  | (l)        | (m)  | (n) | (o)  | (p) |      |  |
| 17.3              | 13.6 | 11.5               | DB                         | 12.7      | 31.6 | 1.9        | 1.2                                                | 11.3       | 32.5 | 10.2       | 33.2 | 9.1 | 33.9 | 7.8 | 34.8 |  |
|                   |      |                    | WB                         | 8.0       | 27.6 | 1.6        | 1.1                                                | 6.9        | 28.4 | 6.0        | 29.1 | 5.1 | 29.7 | 4.0 | 30.6 |  |

Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 24.7   | 20.7  | 21.3  | 21.9  | 23.5  | 25.4  | 27.1  | 28.1  | 28.3  | 27.6  | 26.2  | 23.7  | 22.0  |
|                                                                   | DBStd       | 3.19   | 2.51  | 2.38  | 2.16  | 1.59  | 1.29  | 1.14  | 1.07  | 1.04  | 1.08  | 1.47  | 1.74  | 2.41  |
|                                                                   | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                   | HDD18.3     | 21     | 9     | 6     | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 4     |
|                                                                   | CDD10.0     | 5354   | 332   | 318   | 370   | 404   | 476   | 514   | 561   | 566   | 527   | 501   | 412   | 373   |
|                                                                   | CDD18.3     | 2332   | 82    | 89    | 114   | 154   | 218   | 264   | 303   | 308   | 277   | 243   | 162   | 118   |
|                                                                   | CDH23.3     | 19691  | 135   | 171   | 304   | 634   | 1652  | 2867  | 3674  | 3824  | 3189  | 2207  | 702   | 332   |
| Wind                                                              | WSAvg       | 4759   | 1     | 1     | 4     | 12    | 123   | 686   | 1271  | 1395  | 909   | 323   | 31    | 4     |
|                                                                   | WSAvg       | 5.6    | 6.4   | 6.4   | 6.7   | 6.2   | 5.5   | 4.5   | 4.1   | 4.1   | 4.9   | 6.0   | 6.3   | 6.3   |
|                                                                   | PrecAvg     | 1281   | 64    | 85    | 79    | 52    | 98    | 162   | 139   | 159   | 201   | 115   | 60    | 69    |
|                                                                   | PrecMax     | 1768   | 160   | 209   | 223   | 172   | 219   | 441   | 276   | 354   | 425   | 207   | 151   | 233   |
|                                                                   | PrecMin     | 854    | 16    | 5     | 5     | 1     | 0     | 0     | 20    | 32    | 61    | 19    | 6     | 0     |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | PrecStd     | 250    | 43    | 50    | 61    | 45    | 62    | 115   | 79    | 93    | 120   | 59    | 40    | 58    |
|                                                                   | 0.4% DB     | 26.4   | 26.3  | 27.2  | 27.7  | 29.6  | 30.9  | 31.9  | 32.2  | 31.5  | 30.2  | 28.4  | 27.2  | 27.2  |
|                                                                   | 0.4% MCWB   | 23.1   | 23.1  | 24.4  | 24.2  | 26.2  | 26.5  | 27.2  | 27.4  | 26.9  | 26.2  | 24.6  | 23.8  | 23.8  |
|                                                                   | 2% DB       | 25.4   | 25.5  | 26.3  | 27.0  | 28.3  | 30.1  | 31.1  | 31.4  | 30.6  | 29.4  | 27.5  | 26.4  | 26.4  |
|                                                                   | 2% MCWB     | 22.4   | 22.6  | 23.3  | 23.4  | 24.7  | 25.8  | 26.7  | 26.9  | 26.3  | 25.4  | 24.1  | 23.4  | 23.4  |
|                                                                   | 5% DB       | 24.7   | 25.0  | 25.6  | 26.5  | 27.7  | 29.6  | 30.4  | 30.4  | 29.9  | 28.7  | 26.9  | 25.7  | 25.7  |
|                                                                   | 5% MCWB     | 21.9   | 22.2  | 22.5  | 23.1  | 24.0  | 25.4  | 26.1  | 26.1  | 25.7  | 24.7  | 23.8  | 23.1  | 23.1  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 10% DB      | 24.1   | 24.4  | 25.0  | 25.9  | 27.3  | 29.1  | 29.9  | 29.9  | 29.3  | 28.2  | 26.3  | 25.1  | 25.1  |
|                                                                   | 10% MCWB    | 21.5   | 21.7  | 21.9  | 22.7  | 23.6  | 25.1  | 25.6  | 25.6  | 25.3  | 24.4  | 23.2  | 22.6  | 22.6  |
|                                                                   | 0.4% WB     | 24.3   | 24.5  | 25.6  | 25.9  | 27.1  | 27.5  | 28.1  | 28.2  | 27.7  | 27.5  | 26.2  | 25.0  | 25.0  |
|                                                                   | 0.4% MCDB   | 25.5   | 25.6  | 26.6  | 26.5  | 29.7  | 30.2  | 31.6  | 32.0  | 30.9  | 29.5  | 27.3  | 26.4  | 26.4  |
|                                                                   | 2% WB       | 23.4   | 23.6  | 24.6  | 25.1  | 26.7  | 27.4  | 27.7  | 27.7  | 27.1  | 26.8  | 25.2  | 24.1  | 24.1  |
|                                                                   | 2% MCDB     | 24.9   | 25.1  | 26.0  | 26.1  | 28.2  | 29.7  | 30.9  | 31.5  | 30.4  | 28.6  | 26.7  | 25.9  | 25.9  |
|                                                                   | 5% WB       | 22.7   | 22.7  | 23.5  | 24.3  | 24.9  | 26.2  | 26.8  | 27.0  | 26.5  | 26.0  | 24.6  | 23.6  | 23.6  |
| Mean Daily Temperature Range                                      | 5% MCDB     | 24.4   | 24.6  | 25.4  | 25.8  | 27.5  | 29.1  | 30.3  | 30.8  | 29.7  | 28.2  | 26.4  | 25.4  | 25.4  |
|                                                                   | 10% WB      | 21.9   | 22.0  | 22.6  | 23.5  | 24.2  | 25.7  | 26.3  | 26.4  | 26.0  | 25.3  | 24.0  | 23.0  | 23.0  |
|                                                                   | 10% MCDB    | 23.9   | 23.9  | 24.8  | 25.6  | 27.0  | 28.6  | 29.7  | 29.9  | 28.9  | 27.8  | 26.1  | 24.9  | 24.9  |
|                                                                   | 5% DB       | 3.2    | 3.2   | 3.1   | 3.0   | 2.9   | 3.2   | 3.3   | 3.2   | 3.1   | 2.5   | 2.6   | 2.9   | 2.9   |
|                                                                   | 5% WB       | 4.1    | 3.6   | 3.8   | 3.7   | 3.5   | 3.9   | 4.1   | 3.9   | 3.8   | 3.5   | 3.6   | 3.7   | 3.7   |
| Clear-Sky Solar Irradiance                                        | taub        | 0.360  | 0.374 | 0.384 | 0.405 | 0.430 | 0.454 | 0.461 | 0.451 | 0.422 | 0.406 | 0.375 | 0.365 | 0.365 |
|                                                                   | taud        | 2.550  | 2.518 | 2.477 | 2.396 | 2.369 | 2.373 | 2.358 | 2.406 | 2.478 | 2.489 | 2.528 | 2.562 | 2.562 |
|                                                                   | Ebn at Noon | 891    | 903   | 910   | 895   | 867   | 840   | 833   | 842   | 862   | 858   | 866   | 869   | 869   |
|                                                                   | Edh at Noon | 90     | 100   | 109   | 121   | 124   | 123   | 125   | 118   | 108   | 101   | 91    | 86    | 86    |
| All-Sky Solar Radiation                                           | RadAvg      | 3.54   | 4.36  | 5.45  | 6.30  | 6.52  | 6.17  | 6.12  | 5.92  | 5.12  | 4.55  | 3.79  | 3.29  | 3.29  |
|                                                                   | RadStd      | 0.28   | 0.31  | 0.32  | 0.29  | 0.53  | 0.54  | 0.37  | 0.35  | 0.32  | 0.36  | 0.20  | 0.20  | 0.20  |

Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (d)                     | (e)   | (f)     | (g)    | (h)     | (i)   | (j)   | (k)         | (l)     | (m)     | (n)     |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (15 neighbors) | +0.38 | +0.82   | N/A    | N/A     | N/A   | N/A   | N/A         | -18     | +129    | +102    |

Nomenclature: See separate page